

Weekly CT - 04 | Answer Sheet

Cambridge International AS and A Level Computer Science 9618

Data representation | Networks | Hardware and memory

Time: 40 minutes		Total: 40
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1. Data representation [7]

(a) $11010110_2 = 214_{10}$. [1]

(b) $9E_{16} = (9 \times 16) + 14 = 158_{10}$. Award 1 mark for correct method and 1 mark for the final answer. [2]

(c) $+37 = 00100101$. Invert the bits and add 1: $11011010 + 1 = 11011011$, so $-37 = 11011011$. Award 1 mark for the correct two's-complement method and 1 mark for the final 8-bit answer. [2]

(d) Any one valid reason, e.g. hexadecimal is more compact/easier to read than binary; each hex digit represents exactly 4 bits; it reduces transcription errors when reading long binary values. [1]

(e) 128 characters (2^7). [1]

2. Multimedia representation and compression [7]

(a) $1024 \times 512 = 524,288$ pixels. [1] $524,288 \times 24 = 12,582,912$ bits = 1,572,864 bytes. [1] $1,572,864 / 1,048,576 = 1.50$ MiB. [1] [3]

(b) Sample rate is the number of samples taken from the analogue sound wave per second. [1]

(c) Award 1 mark for each effect: (i) more samples / smaller time gaps, so the digital sound more closely represents the original; (ii) more data must be stored, so file size increases. [2]

(d) Run-length encoding (RLE). Accept another valid lossless bitmap compression method if supported by the taught material. [1]

3. Networks and network devices

[8]

(a) Award 1 mark for each valid feature, up to 2: all computers have equal status; each computer can provide/access resources; data/resources are distributed; there is no dedicated central server; each computer is responsible for its own security. [2]

No.	Description	Device
1	Forwards packets between different IP networks using destination IP addresses.	Router
2	Connects devices within a LAN and forwards frames to the correct port using MAC addresses.	Switch
3	Converts/modulates and demodulates signals so a network can communicate over an ISP access line.	Modem
4	Sends incoming data out to all other ports rather than selecting one destination port.	Hub

Award 1 mark for each correct device. [4]

(c) Accept wired or wireless if the justification is technically appropriate. A typical answer is wired because it provides higher bandwidth/faster transfer, lower latency, a more stable connection and less interference. Award 1 mark for a clear choice and 1 mark for one valid justification. [2]

4. Internet, IP addressing and cloud

[8]

computing

(a) Public IP address: visible/routable on the Internet. Private IP address: used within a local network and not directly routable on the public Internet. [2]

(b) Static IP address does not normally change each time the device connects. Dynamic IP address can be allocated/reallocated and may change between connections. [2]

(c) /26 subnet mask = 255.255.255.192. Network address = 192.168.10.64. [2]

(d) Benefit: e.g. accessible from different locations with Internet access; reduced need for local storage; easy sharing/collaboration; provider may handle backup. Drawback: e.g. requires Internet access; third-party dependency; security/privacy concerns; possible ongoing cost. Award 1 mark each. [2]

5. Memory, RAM and ROM [4]

(a) Volatile memory loses its contents when power is removed. [1]

(b) Any one valid difference, e.g. RAM is volatile and read/write working memory, while ROM is non-volatile and stores instructions/data that persist when power is off. [1]

(c) RAM: stores temporary data such as the selected wash program, sensor readings or current program state.

ROM: stores start-up instructions / fixed program instructions for the washing cycles. Award 1 mark each. [2]

6. Types of RAM and types of ROM [5]

(a) Award 1 mark per valid paired difference, up to 2: SRAM does not require refreshing whereas DRAM requires periodic refreshing; SRAM is generally faster whereas DRAM is slower; SRAM is generally more expensive / lower density whereas DRAM is cheaper / higher density. [2]

No.	Description	ROM type
1	Can be programmed once by the user and cannot then be erased.	PROM
2	Can be erased using ultraviolet light and then reprogrammed.	EPROM
3	Can be erased and reprogrammed electrically without removing the chip.	EEPROM

Award 1 mark for each correct type. [3]

7. Atleast 2 hours of mindful study is [0.5]

expected

Award 0.5 mark for acknowledgement / response. [0.5]

8. Thank you again

[0.5]

Award 0.5 mark for acknowledgement / response. [0.5]

Total = 40 marks